



EGH404 Generative AI for Data Generation and Assignment 2 Help Session

Faculty of Engineering

EGH404

Demonstration: Using GenAI for Locating Public Datasets

I need help locating publicly available tabular data on the topic of evaluating the structural performance of recycled concrete. Can you provide me with some datasets, a brief description, and URLs? The data should strictly be tabular. Do not hallucinate and provide links that do not exist.



Below are **publicly available, verifiable tabular datasets** that are relevant to **evaluating the structural/mechanical performance of recycled concrete**. I have **only included datasets that clearly provide structured tabular data (Excel/CSV-style)** and **explicit download links from well-known data repositories**. I have not inferred or fabricated any links.

1. Mechanical Properties of Recycled Aggregate Concrete (RAC)

Repository: Mendeley Data

Format: Excel (.xlsx) – strictly tabular

Demonstration: Generating Synthetic Data with MS CoPilot

Refer to the full prompt
provided in the practical
page

```
1 You are a data generator. Create a synthetic dataset in **CSV format only** (no
2 prose, no markdown tables, no code fences).
3 1) ROW COUNT:
4 - Generate approximately 200 rows (acceptable range: 190-220).
5
6 2) SCHEMA:
7 - Columns (in order): PlantID, PlantName, Country, Region, EnergySource,
8 CapacityMW, CommissionDate, OperatorID, OperatorName, AnnualGenerationMWh,
9 CO2SavingsTonnes
10
11 3) RELATIONSHIPS:
12 - PlantID: unique
13 - OperatorID → OperatorName (one-to-one)
14 - PlantID → PlantName, Country, Region, EnergySource, CapacityMW, CommissionDate
15 (functional dependency)
16 - EnergySource ∈ {Solar, Wind, Hydro, Biomass, Geothermal}
17 - CapacityMW depends on EnergySource:
18     - Solar: 5-150 MW
19     - Wind: 10-300 MW
```



EGH404 – Assignment 2 Help Session

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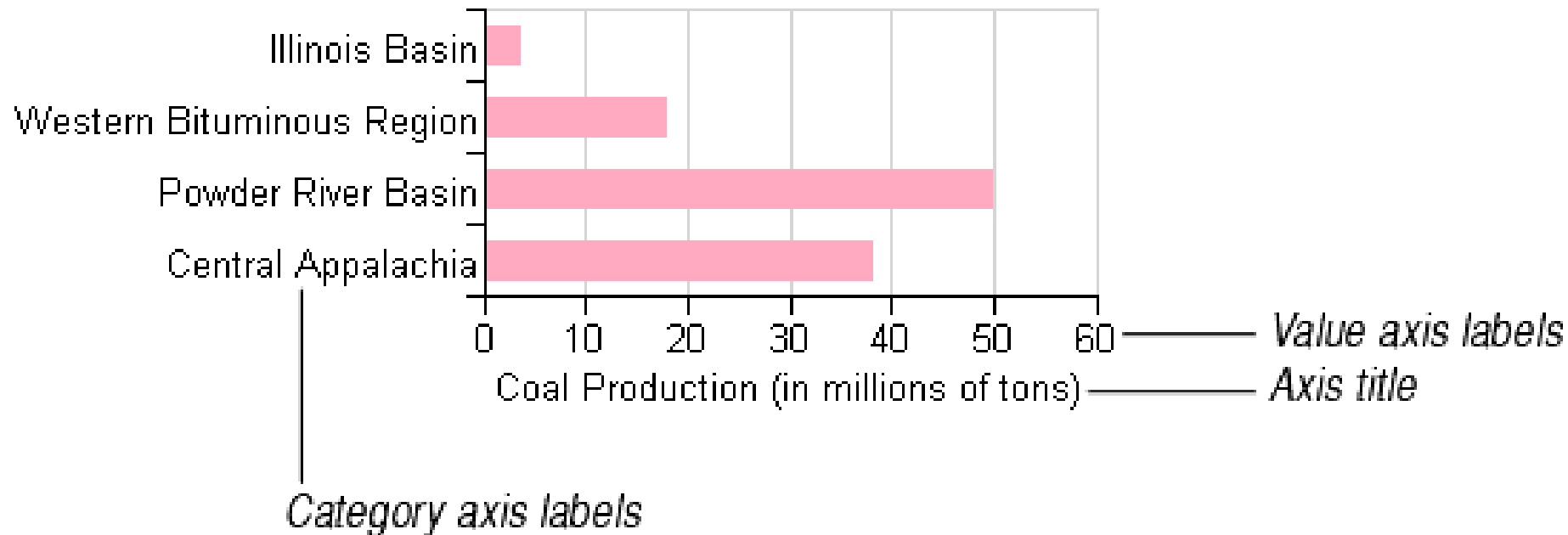
EGH404

Plots

Criteria	Ratings		Pts
S1. T1 (a) Create a single box plot visualisation to identify outliers across daily bikeway usage data.	1 Pts Full marks A single, well-formatted box plot is created with correct data	0 Pts No marks Box plot is missing, incomplete, or does not include correct data	1 pts

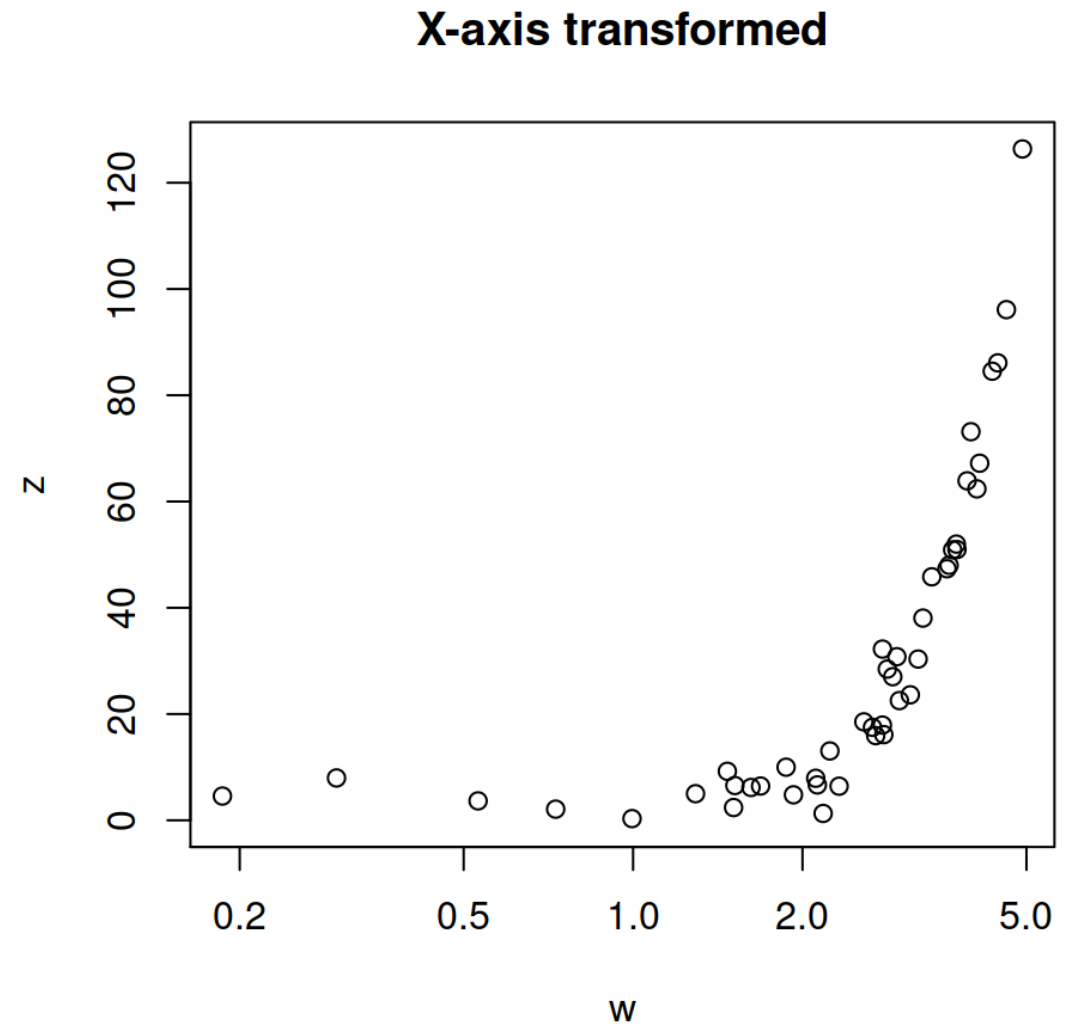
Plots

Should have a plot title, axis titles, consistent axis labels, and a legend (if applicable).



Should have a plot title, axis titles, consistent axis labels, and a legend (if applicable).

Plots



SECTION 1 – DATA WRANGLING (1.1.C & 1.2.C)

S1. T1. (c) Identify problematic data (e.g., outliers, missing data, errors) and describe how you will handle them in your cleaned dataset	4 Pts Full marks (i) Clearly identifies all problematic data points across all three weather factors, including specific dates. (ii) Provides a structured table detailing issue type, date, and justification for chosen cleaning method based on statistical or domain reasoning.	2 Pts Partial Marks Several problematic data points identified, but explanation or justification is incomplete or lacks clarity.	0 Pts No marks Only a few problematic data points identified, or plan for handling them is vague.	4 pts
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SECTION 1 – DATA WRANGLING (1.1. C & 1.2.C)

Name of the column with the problematic data, type of the problem (outliers, missing data, errors) and the value.	Specify the date with problematic data	How do you intend to handle the problematic data identified?	Justification

Detailed Description:

- What is the best outlier/missing data handling mechanism for this data?
- Don't just write removing outliers
- For missing data imputation explain why this was chosen.
- E.g. Why choose mean/median/mode to impute

SECTION 1 – DATA WRANGLING (1.1. C & 1.2.C)

Name of the column with the problematic data, type of the problem (outliers, missing data, errors) and the value.	Specify the date with problematic data	How do you intend to handle the problematic data identified?	Justification

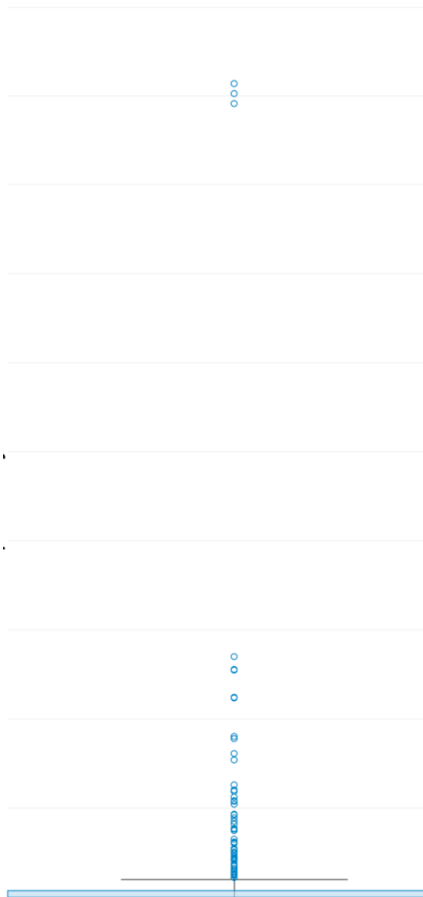
Use up to 3 examples for each **problem type** within **each bikeway**.

- Outliers
- Missing data
- Errors

- Bicentennial Bikeway
- North Brisbane Bikeway
- No Need to split between pedestrians and cyclists

If more than three **instances** exist, report only three. If fewer than three exist, report all that you can find.

Interesting Cases



- Outliers: does not always mean that you need to remove them.
- First consider whether the behavior in the data is explainable.
 - In addition to the box plots, take a look at the line plots to see if there are consistent patterns.
 - Check if there are any signs of errors in measurements or sensors, like consistent readings over several days or days that don't follow the overall data trend.
 - You can also search online for any events (e.g., floods, bikeway closures, etc.) that might explain those inconsistencies.
- Based on this, you can make an informed decision on whether or not to keep the data points in your analysis.

Supporting Evidence

- Source code (for MATLAB or Python users),
- Screenshots of Excel formulas, and screenshots of Excel “Data Analysis” tools
- Please include your supporting evidence in the Appendix (**copy and paste the code into the document, or if using live scripts merge the document PDF with the exported live script PDF**).

Supporting Evidence - MATLAB

```
% Read the CSV file into a table
```

```
bike_ped_data_table = readtable('bikeway_data.csv');
```

Warning: Column headers from the file were modified to make them valid MATLAB identifiers before creating variable names for the table. The original column headers are saved in the VariableDescriptions property.
Set 'VariableNamingRule' to 'preserve' to use the original column headers as table variable names.

```
% Display the first few rows of the table to understand its structure
```

```
disp(bike_ped_data_table(1:5, :))
```

Date	A001Pedestrian	A001Cyclist	A002Pedestrian	A002Cyclist	A023Pedestr
{'Sat, 01 01 22'}	NaN	NaN	698	38	2581
{'Sun, 02 01 22'}	NaN	NaN	591	156	4426
{'Mon, 03 01 22'}	NaN	NaN	614	152	3641
{'Tue, 04 01 22'}	NaN	NaN	514	158	2664
{'Wed, 05 01 22'}	NaN	NaN	454	111	2604

```
% Isolate data for A004 and A007
```

```
bike_columns_of_interest = {'A004NPedestrian', 'A004ECyclist',  
                             'A007Pedestrian', 'A007Cyclist'};
```

```
% Change date format
```

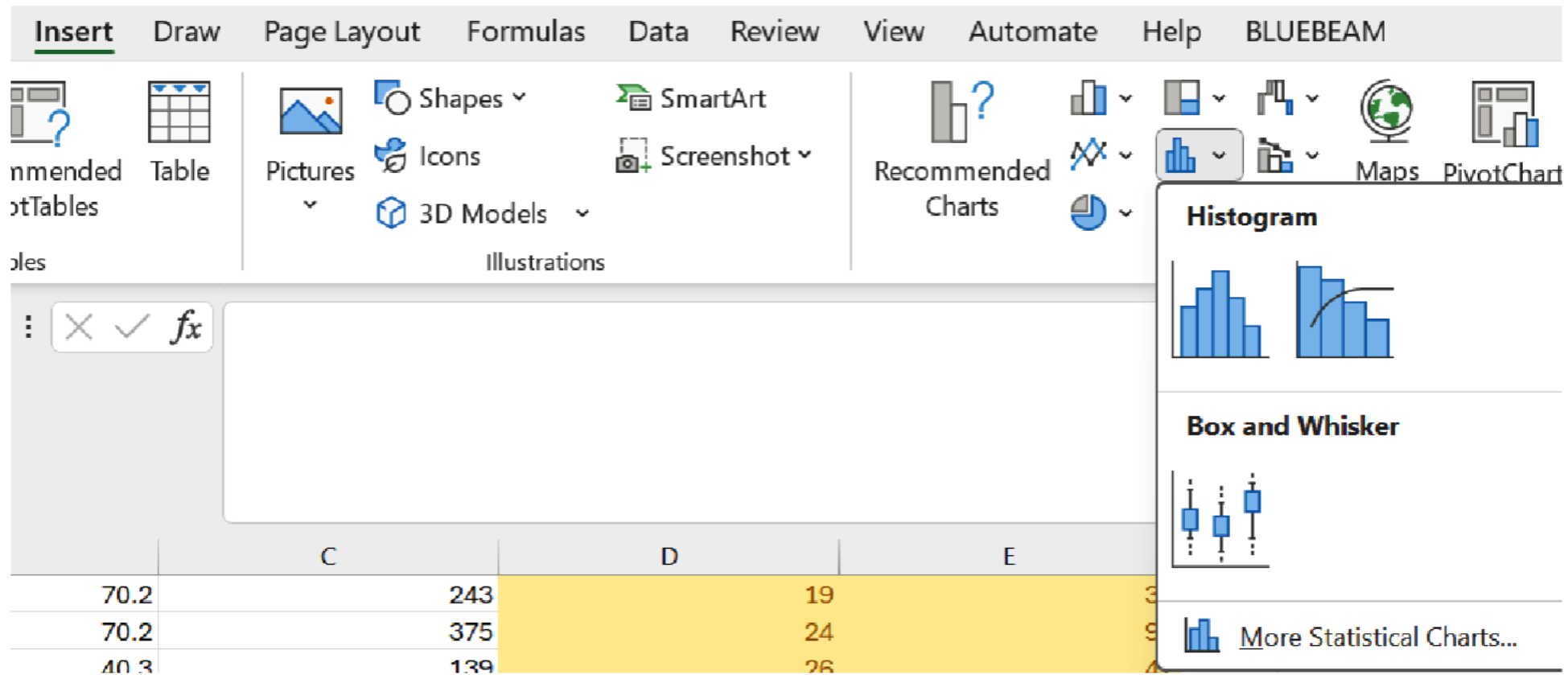
```
% Extract data and form table
```

```
bikeway_data = bike_ped_data_table(:, ['Date' bike_columns_of_interest]);
```

```
% Display the isolated data
```

```
%disp(bikeway_data);
```

Supporting Evidence - Excel



Supporting Evidence - Excel

Select Data Source

Chart data range: ='Q1.1a (2)!\$B\$2:\$E\$366

Switch Row/Column

Legend Entries (Series)

Add Edit Remove ^ v

Eleanor Schonell Cyclists

Eleanor Schonell Pedestrians

Jack Pesch Cyclists

Jack Pesch Pedestrians

Horizontal (Category) Axis Labels

Edit

Hidden and Empty Cells

OK Cancel

Supporting Evidence - Excel

The screenshot shows the Microsoft Excel interface with the 'Conditional Formatting' menu open. The ribbon includes File, Home, Insert, Draw, Page Layout, Formulas, Data, Review, View, Automate, Help, and BLUEBEAM. The 'Conditional Formatting' dropdown menu is expanded, showing options: Highlight Cells Rules, Top/Bottom Rules, Data Bars, Color Scales, Icon Sets, New Rule..., Clear Rules, and Manage Rules... On the right side of the menu, there are additional options: Greater Than..., Less Than..., Between..., Equal To..., Text that Contains..., A Date Occurring..., and Duplicate Values... The formula bar shows 'C239' and the value '3241'. The spreadsheet grid shows data for dates from 8/08/2022 to 13/08/2022 across columns A to E.

	A	B	C	D	E
221	8/08/2022	1301.3	1946	385	
222	9/08/2022	1396.85	2044	443	
223	10/08/2022	960.05	1246	763	
224	11/08/2022	1418.95	2313	455	
225	12/08/2022	1243.45	1757	526	
226	13/08/2022	293.15	691	159	

25/08/2022	234	3796
26/08/2022	249	3241
27/08/2022	204.5	1612

Use the correct version of the dataset

You are required to:

1. Review the unit content on data wrangling, data visualisation, statistics, correlation and regression, and Generative AI for data generation. While completing your Assessment 2, be mindful of its Rubric and stated criteria for marking of this assessment.
2. Download and use the [brisbane bikeway counters.csv](#) ↓ and [brisbane weather.csv](#) ↓ datasets for answering the questions in Section 1.
3. Download and use the [brisbane bikeway counters cleaned.csv](#) ↓ and [brisbane weather cleaned.csv](#) ↓ datasets for answering the questions in Sections 2 and 3.

SECTION 2 – CORRELATION ANALYSIS USING REAL-WORLD DATA

S2.T1: Correlation Coefficient

	Correlation value
Eleanor Schonell Cyclists and Ped	Command
Jack Pesch Cyclists and Ped	
Cyclists on ES and JP	
Pedestrians on ES and JP	

	Correlation value
Eleanor Schonell Cyclists and Ped	Values
Jack Pesch Cyclists and Ped	
Cyclists on ES and JP	
Pedestrians on ES and JP	

SECTION 2 – CORRELATION ANALYSIS USING REAL-WORLD DATA

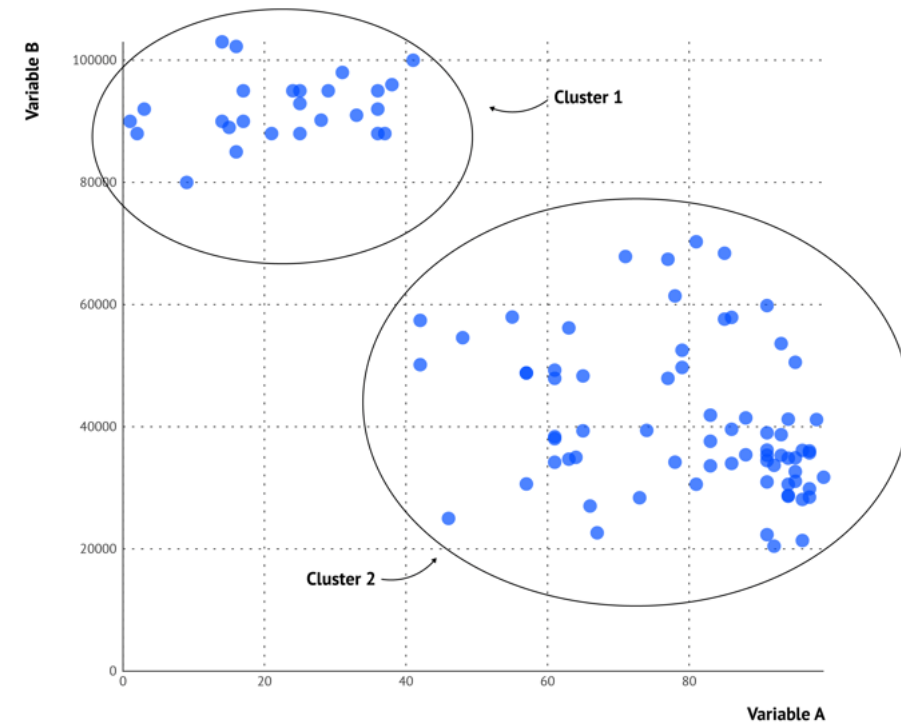
S2.T3: Correlation Analysis Interpretation (400 words)

General observations

- What sort of clusters do you see in the plots?

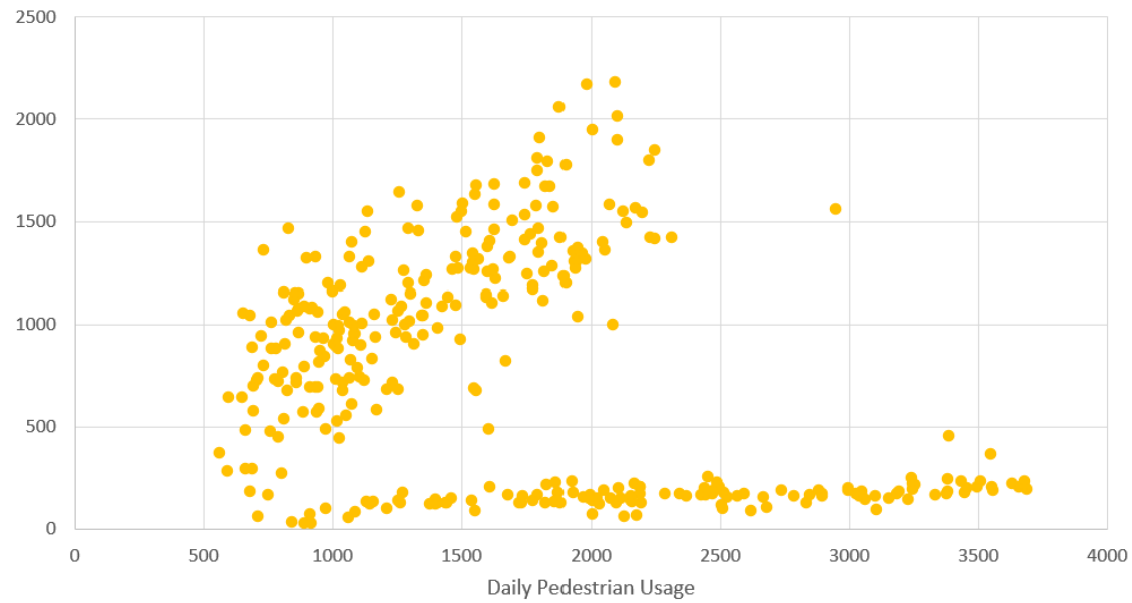
Reasons / explanations/ hypothesis for observations

- What do they imply. Seasonal variations, Demographics and preferences?
- What does the correlation coefficient implies?



Interesting Cases

Pearson Correlation Coefficient = -0.35



SECTION 3 – MULTIPLE LINEAR REGRESSION ANALYSIS

3.1 Multiple Regression Models & Supporting Visualisations

```
J_cyclist_regression =  
Linear regression model:
```

```
[REDACTED]
```

Estimated Coefficients:

	Estimate	SE	tStat	pValue
(Intercept)	[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]
rainfall	[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]
solarexposure	[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]
maxtemp	[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]

Number of observations: [REDACTED] Error degrees of freedom: [REDACTED]

Root Mean Squared Error: [REDACTED]

R-squared: [REDACTED], Adjusted R-Squared: [REDACTED]

F-statistic vs. constant model: [REDACTED], p-value = [REDACTED]

SECTION 3 – MULTIPLE LINEAR REGRESSION ANALYSIS

3.1 Multiple Regression Models & Supporting Visualisations

SECTION 3 – MULTIPLE LINEAR REGRESSION ANALYSIS

Using your cleaned data, please answer the following questions.

To Answer Sections 2 and 3 please use the cleaned datasets (brisbane_bikeway_counters_cleaned.csv, brisbane_weather_cleaned.csv) provided in Canvas Assessment 2 – Details page.

BCC wants to further investigate the relationship how daily weather factors (i.e. daily rainfall, solar exposure, maximum temperature) influence usage Bicentennial Bikeway, Brisbane – and whether there is any difference in how weather influences cyclist versus pedestrian usage of this bikeway. To answer the questions in this section, please split the dataset as follows: use data from January 1st, 2014, to December 31st, 2017 (inclusive) as the training set, and use all remaining data as the testing set.

S3.T1: Multiple Regression Models & Supporting Visualisations

Please conduct Multiple Regression analyses to investigate if one or more daily weather factors (i.e. rainfall, solar exposure, maximum temperature) influence **pedestrian usage on the Bicentennial Bikeway, Brisbane**. Do not create separate regression models for each independent variable. Instead, build a single regression model that includes all the identified independent variables. You may need to repeat the regression modelling process until all variables have statistically significant p-values.

- Provide model summary table (including P-values, R^2 , adjusted R^2 , RMSE on train and test sets)
- Generate a scatter plot illustrating the observed data points for the independent variables against the dependent variable and overlay the fitted regression line to demonstrate the model's representation of the relationship.
- Discuss Any model simplification (i.e. removal of independent variables from the model) with appropriate justification. Discuss the impact of simplifying the model on performance if any.

SECTION 3 – MULTIPLE LINEAR REGRESSION ANALYSIS

3.1 Multiple Regression Models & Supporting Visualisations

- **Regression summary table** should be generated on the **training data**.
- **RMSE** should be reported on **both train and test splits**.
- **Scatter plot** → Training data

SECTION 3 – MULTIPLE LINEAR REGRESSION ANALYSIS

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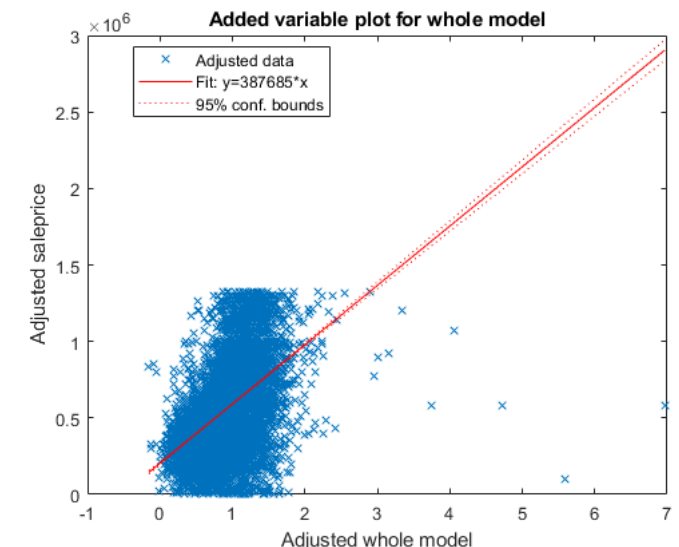
SECTION 3 – MULTIPLE LINEAR REGRESSION ANALYSIS

3.2 Multiple Regression Interpretation (400 words)

S3. T2 Interpret the results from the two Multiple Regression Models	2 Pts Full marks Provides a clear, well-structured interpretation of both regression models. Uses appropriate regression outputs to justify conclusions.	1 Pts Half Marks Interpretation is present but lacks depth, clarity, or sufficient reference to regression outputs. May not fully address both models.	0 Pts No marks No interpretation provided or interpretation is incorrect and unsupported by regression outputs.	2 pts
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What do F statistic, R², and p-values tells you about the model?

Use the plots to describe how well the model can describe the trends in the data



SECTION 4 –LOCATING OR SYNTHESISING PRELIMINARY DATA

In This Section, Choose One Pathway (A Or B) — Do Not Complete Both

S4.TA: Locating Real-World Dataset (Publicly Available)

You will locate, and justify, a publicly available real-world tabular dataset that is broadly related to your research topic from Assessment 1. Note that you will use the same dataset during your Assessment 3 – Project Pitch presentation, to demonstrate preliminary evaluation.

OR

S4.TB: Synthetic Data Generation

Using the synthetic data generation procedure introduced during practical session, create a synthetic dataset that aligns with the research question you identified in your response to Assignment 1. You should use the same research topic from Assessment 1 to define the variables for synthetic data generation and for your Assessment 3 – Project Pitch presentation. Ensure the dataset is relevant, appropriately structured, and suitable for subsequent analysis.

SECTION 5 – CORRELATION ANALYSIS USING PRELIMINARY DATA

SECTION 5 – CORRELATION ANALYSIS USING PRELIMINARY DATA

Using the dataset you used in Section 4 (i.e. real-world data you located or the synthetic dataset you generated) answer the following questions.

S5.T1: Data Modelling Using Preliminary Data

- a) In your dataset Identify two or more variables that appear to be related (i.e., feature(s) and a target). Provide scatter plots and correlation matrix.
- b) Describe the nature of their relationship (e.g., linear, inverse, nonlinear, or no clear pattern). Support your answer using outputs generated using relevant data analysis concepts (e.g., correlation coefficient, scatter plots, etc.).
- c) Compare the relationships you observe in your data with those reported in the relevant literature for your chosen topic.

(5 pts)